

Signatures of graphs and cumulants of commutative combinatorial Hopf algebras

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Abstract

This thesis aims to link graph theory and combinatorial Hopf algebras through signatures. Both theories comprise combinatorial structures practical in the modelling or solving of non-linear processes. For instance, polynomials described on a signature reduce to combinatorial computations. However, they have complementary characteristics. While signatures are typically defined for time-ordered data, the usefulness of graph modelling rests on the lack of a sequential order. This lack of order complicates other tasks, such as comparing whether two graphs are similar.

This research applies signatures to the graph-comparison problem. Conversely, it draws the notion of connectivity from graph theory to solve a problem for signatures. We accompany concepts and results with concrete examples throughout the text.

In the first part, several signatures for graphs are proposed. These are infinite-dimensional vectors with a term-wise multiplication and a vector-wise convolution, which we register with a combinatorial Hopf algebra. Each vector arises from counting different substructures within a graph, such as homomorphisms or subgraphs. Accordingly, non-isomorphic graphs correspond to distinct vectors. We first uncover Hopf algebras compatible with each signature. The second question is whether such algebras are isomorphic or not. Lastly, and most importantly, the goal is to find transformations between signatures that respect the Hopf algebra structure, i.e. product and convolution.

Following the steps of other research in both computer science and combinatorial Hopf algebras, we multiply the entries of these counting vectors. Then, they are expressed as the linear sum of other entries of the same vector. This calculation allows us to define bilinear products of graphs, which we dualise into coproducts. We later pair product-coproduct and verify when the coproduct is multiplicative and the existence of a particular operator, hence forming Hopf algebras. These Hopf algebras are all commutative, cocommutative, and conilpotent. Their coproducts cannot decompose connected graphs, so they form what we call the space of primitive elements. Using Cartier's theorem, all these Hopf algebras are isomorphic. We then develop Cartier's theorem into several tools for signatures. These extend linear maps defined in the space of primitive elements into (co)multiplicative maps.

By Cartier's theorem, we confirmed that our signatures for graphs are all equivalent. While this was already known as combinatorial functions, the classical proofs are typically done pairwise. In this thesis, we provide a unifying algebraic explanation of such equivalency. Namely, the information in connected graphs is complete and irreducible (within the finite-simple-graph setting). Completeness is well-known.

Irreducibility was widely expected but now receives a combinatorial Hopf algebraic proof.

The second part computes the algebraic cumulants for quasishuffle (Hopf) algebras. In time-series theory, Iterated Integral Signatures (IIS) resp. Iterated Sums Signatures (ISS) have been proved compatible with shuffle resp. quasishuffle Hopf algebras. Evaluating non-linear expressions with the correct product yields a linear expression. The question driving the research was simplifying the evaluation of a signature on a monomial of another product. For instance, computing IIS in quasishuffle monomials or ISS in shuffle monomials. One can use algebraic cumulants for any pair of commutative graded algebras, such as quasishuffle/shuffle. These cumulants turn quasishuffle monomials into shuffle polynomials using the incidence algebra of set partitions. The quest is to find cancellation-free expressions for these cumulants. And so we do for the reversed cumulants, i.e. shuffle monomials in terms of quasishuffle polynomials.

We interpret each term of a quasishuffle monomial using a table where each row represents a factor word, each column represents one letter in the final word, and each entry indicates from which words that final letter is computed. Due to the associativity of the product, we noticed that each term appears when evaluating submonomials and combining them afterwards. The rows of the tables then obey an extraction-contraction decomposition. Originally a concept from numerical analysis, the last decade has seen the rise of extraction-contraction Hopf algebras to explain Tutte's deletion-contraction recursion on (hyper)graphs. We borrow from these constructions. Furthermore, each table is assigned to a graph, where the vertices are the rows of the table. When that graph is connected, we say that the table is connected.

We later prove that connected tables precisely describe the algebraic cumulants. The cancellation-free expression of the reverse cumulants requires a graph's lattice of connected partitions. We also prove that any commutative Hopf algebra with a deconcatenation coproduct must also satisfy an extraction-contraction decomposition. Accordingly, their cumulants are given by connected tables.

The two parts of this thesis develop abstract tools and apply them to concrete cases. In both graph and signatures theory, non-linear expressions are translated into combinatorial identities that boil down to connected graphs. Emphasis is placed on all the (Hopf) algebras used in this study as being commutative and the graphs undirected.

Keywords: Hopf algebra, combinatorial signatures, graph counting, cumulants, shuffle and quasishuffle products, homomorphism numbers, extraction-contraction, Chen's identity.

Contents

Abstract	iii
Acknowledgements	v
1 Introduction	1
1.1 Scope, context, and setting	2
1.2 Signatures for graphs	5
1.3 Shuffle cumulants	7
1.4 Structure of the Thesis	9
2 Classical combinatorial Hopf algebras	11
2.1 Combinatorial Hopf algebras	11
2.1.1 Graded vector spaces	11
2.1.2 Algebras	13
2.1.3 Coalgebras	14
2.1.4 Bialgebras and Hopf algebras	16
2.1.5 Dualising a Hopf algebra	18
2.1.6 Characters and infinitesimal characters	20
2.2 The symmetric Hopf algebra of graphs	21
2.2.1 Background on graph theory	21
2.2.2 The Hopf algebra generated by connected graphs	23
2.2.3 Other Hopf algebras of graphs	24
2.3 Algebraic cumulants	24
3 Graph signatures	29
3.1 Different types of injective homomorphisms	31
3.2 Hopf algebras of graphs	32
3.2.1 A finite-type graded vector space on graphs	32
3.2.2 Section coefficients for graphs	32
3.2.3 (Co)algebras of graphs	35
3.2.4 Bialgebras and Hopf algebras	37
3.3 Isomorphisms to the symmetric Hopf algebra	39
3.3.1 Bicommutative unipotent Hopf algebra	39
3.3.2 The symmetric Hopf algebra of graphs	42
3.3.3 Infinitesimal Chen's Identity	43

3.4	Existence of combinatorial signatures for graphs	44
3.4.1	Combinatorial and homomorphism numbers	44
3.4.2	Counting functionals for graphs	45
3.4.3	Signature properties	46
3.5	Transport maps for graph signatures	49
4	Algebraic cumulants of shuffle algebras	55
4.1	Quasi-shuffle connected covers	57
4.1.1	Reshuffle tables for the quasishuffle product	58
4.1.2	Extraction-contraction of covers	60
4.2	Quasi-shuffle cumulants	65
4.2.1	Moment-cumulant formula for quasishuffle monomials	65
4.2.2	Moment-cumulant formula for shuffle monomials	67
4.2.3	Nested cumulants in the cumulant-cumulant formula	70
4.3	Commutative B_∞ -algebras	73
4.3.1	Weighted extraction-contraction formulas	74
4.3.2	The moment-cumulant formula	77
4.4	Further research	78
	Bibliography	81